

Vol 14 Chapter 9 — Kolmogorov Complexity and AC(0)

Keeper (author pass — deep math/physics revision)

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Chapter 9 — Kolmogorov Complexity and AC(0)

Level 1 — one sentence

Kolmogorov complexity $K(x)$ is the length of the shortest program producing string x on a fixed universal Turing machine, providing the absolute measure of “compressibility”; AC(0) is the complexity class of constant-depth polynomial-size circuit families (bounded-depth counting); BST identifies the substrate’s K-type ladder $\{rank, N_c, n_C, C_2, g\}$ as the Kolmogorov-minimum specification of observable physics, with substrate “computation” inherently AC(0).

Level 2 — graduate-physicist precision

9.1 Kolmogorov complexity

For string x :

$$K(x) = \min\{|p| : U(p) = x\}$$

with U universal Turing machine, $|p|$ length of program p .

Properties: - $K(x) \leq |x| + O(1)$ - Incompressibility: most strings have $K(x) \geq |x| - O(\log |x|)$ - Uncomputable: K is not recursive

9.2 BST as Kolmogorov-minimum (computed per Toy 3519)

BST claim (I-tier, encoding-scheme-dependent — K is uncomputable in general, so this is a side-by-side encoding comparison, not a Kolmogorov-complexity theorem):

$$K(\text{all observable physics}) \approx K(\{\text{rank} = 2, N_c = 3, n_C = 5, C_2 = 6, g = 7\} + D_{IV}^5) \approx 73 \text{ bits}$$

per Elie Toy 3519 computation (Sunday 2026-05-24), updated from the earlier ~100 bit estimate.

Compare (under canonical encoding assumption): - Standard Model with 19-27 free parameters: $K \geq 19 \cdot 32 = 608$ bits - BST: $K \approx 73$ bits (D_IV⁵ structure + 5 primaries minimum encoding)

BST achieves ~ 8x compression vs Standard Model under this encoding scheme.

If the estimate holds across encoding schemes: BST is materially more compressible than the Standard Model. Caveat: tight bound requires choosing canonical encoding for D_IV⁵ structure; Toy 3519 is computational best-effort, not provable lower bound.

9.3 AC(0): bounded-depth counting

AC^0 = languages accepted by polynomial-size, constant-depth circuit families using unbounded-fan-in AND, OR, NOT gates.

Functions NOT in AC^0 : parity (Hastad 1986), majority (Razborov-Smolensky 1987), reachability in graphs (Furst-Saxe-Sipser). Functions IN AC^0 : addition, fixed-precision multiplication, most polynomial-time-decidable trivialities.

Casey's standing skill /ac0: enforces AC(0)-thinking — reduce everything to counting at bounded depth before invoking sophisticated machinery.

9.4 BST substrate AC(0)

BST hypothesis: substrate computation is AC(0) — bounded-depth ($\leq$ depth 1 or 2 under Casey strict-protocol; T421+T316), polynomial-size circuit family.

Implications: - Substrate cannot compute parity (per Hastad lower bound) - Substrate cannot compute majority (per Razborov-Smolensky) - Substrate restricted to BST-primary counting operations

Evidence: every BST-derived constant reduces to bounded-depth counting (heat kernel coefficients, partition functions, etc.).

9.5 $P \neq NP$ via curvature

BST identifies $P \neq NP$ with substrate-K-type non-linearizability (Casey's Curvature Principle, memory). The substrate K-type space has Gauss-Bonnet curvature obstructing linearization.

Vol 14 Ch 7 AC graph supports this: substrate's path-finding (NP-like search) requires curvature-respecting algorithms.

9.6 K-audit anchors

- T421: depth ≤ 1 under Casey strict
- T316: depth $\leq$ rank = 2
- Casey's Curvature Principle (memory)
- /ac0 skill (Casey standing tool)

Level 3 — 5th-grader accessibility

Kolmogorov complexity $K(x)$ = shortest program producing x . Most strings are incompressible. BST has $K \approx 73$ bits for all observable physics (Elie Toy 3519) vs ≥ 608 bits for Standard Model — 8x compression. $AC(0)$ = constant-depth circuits; can't compute parity or majority (Hastad, Razborov-Smolensky). BST hypothesis: substrate is AC(0) — bounded-depth counting only. Casey's /ac0 skill: reduce to counting first, sophistication later. $P \neq NP$ via curvature: substrate K-type curvature prevents linearization.

What comes next

Chapter 10 develops substrate complexity classes.

Where to look this up

- Kolmogorov 1965; Chaitin 1966
- Hastad 1986 (AC(0) parity)
- BST: T421, T316; Casey's Curvature Principle